Class 9

THE FUNDAMENTAL UNIT OF LIFE

Ch 5 | Class 9th (Science)

Handwritten Notes

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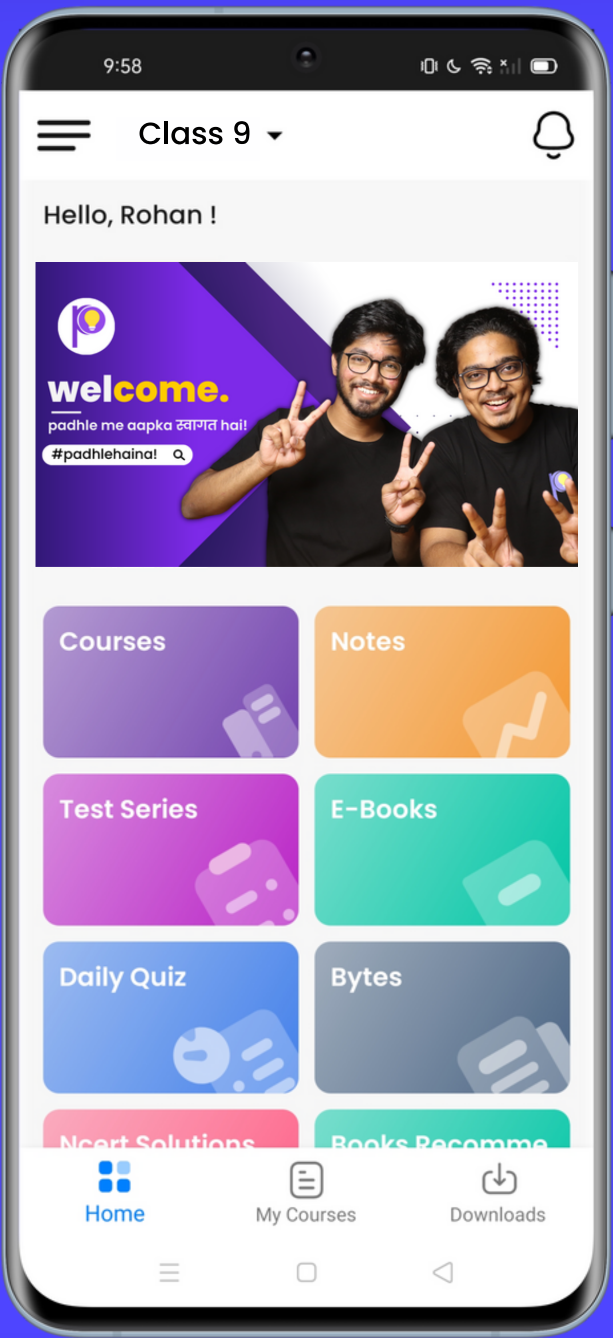

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FUNDAMENTAL UNIT OF LIFE

* What are living organisms made of ?

- The Basic, Structural and Functional Units of life is Cell.

Discovered by Robert Hook in 1665

* Cell Theory

Matthias Schleiden
1838

- All plants are made of cells.

Theodore Schwann
1839

- All animals are made of cells.

Rudolf Virchow
1855

- All cells come from pre-existing cells.

- All living things are composed of cells.
- Cells are the basic structural and functional units.
- New cells are produced from existing cells.

* Types of Organisms

Unicellular

- Made of only one cell.
- Single cell performs all life functions. (move, eat, reproduce)
- Example : bacteria, amoeba etc.

Multicellular

- Made of multiple cells.
- Different cells specialize
- Example : humans

* Types of Cells

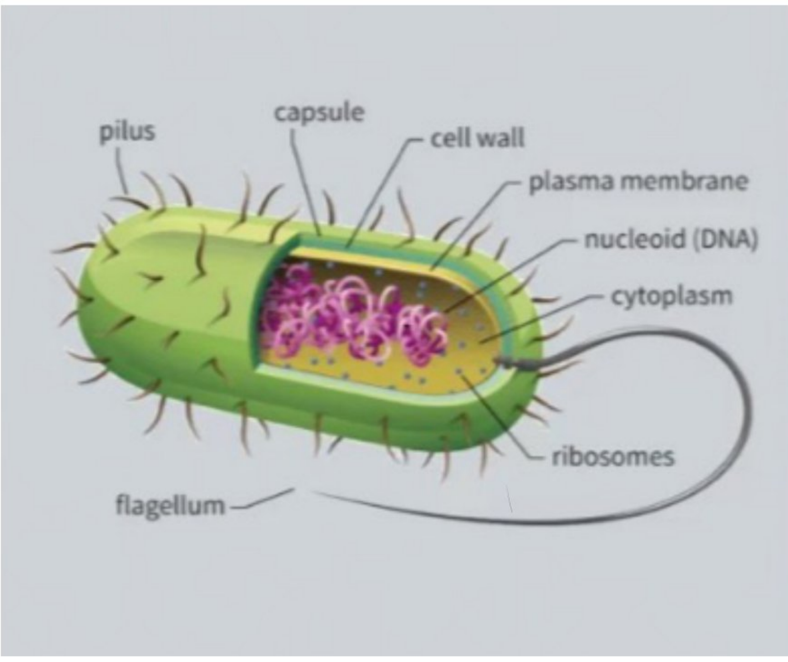
Prokaryotic

- Size (1-10 micro meter).
- One chromosome present.
- The nucleus is not well defined.

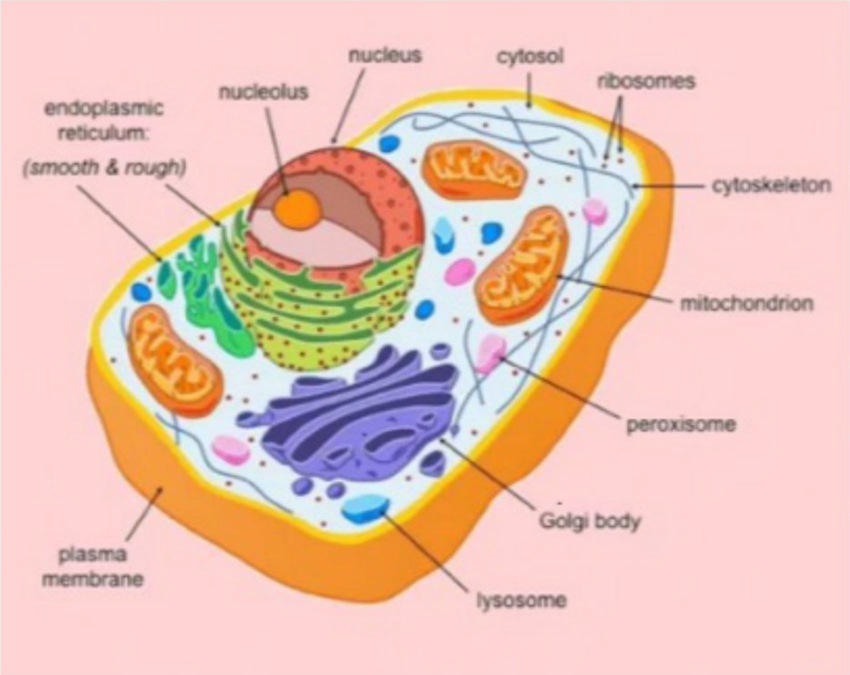
Eukaryotic

- Size (5-100 micro meter)
- more than one chromosome.
- The nucleus is well defined and surrounded by nuclear membrane.

* Prokaryotic Cell

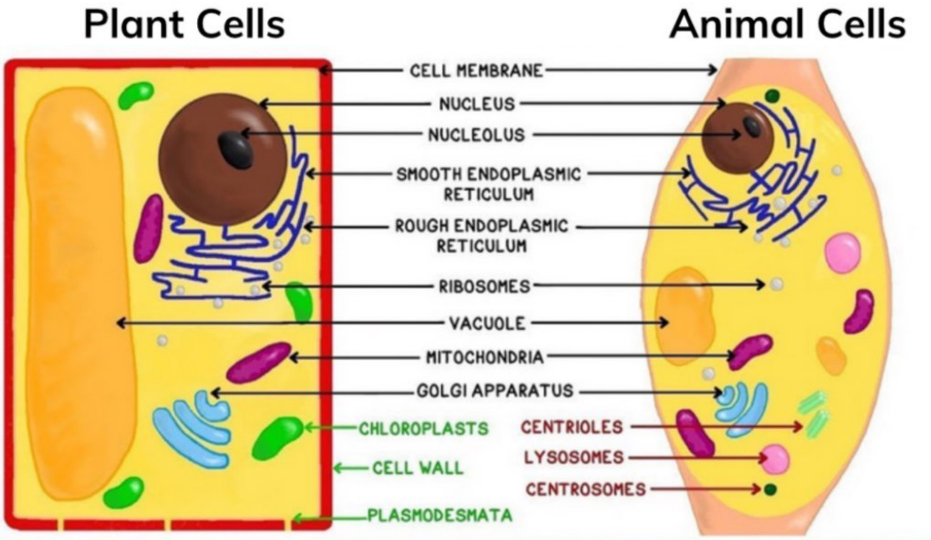


★ Eukaryotic Cell

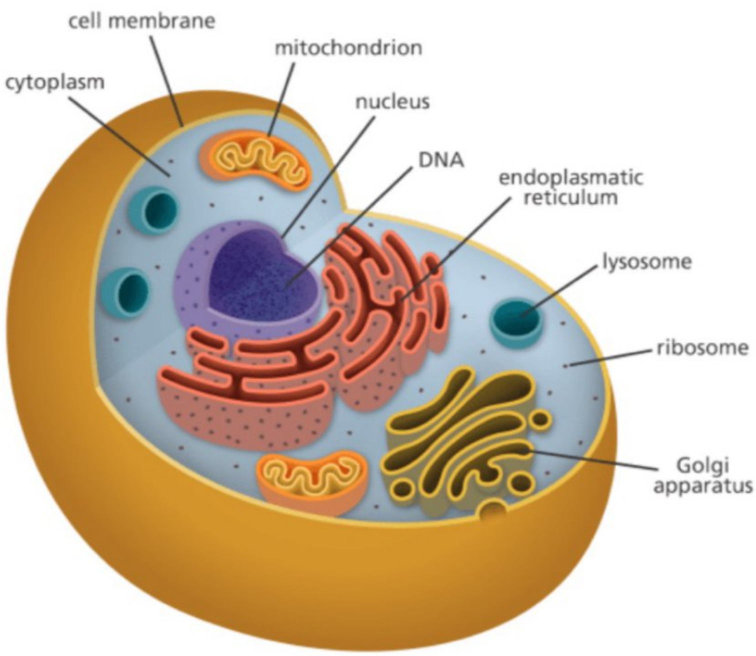


Characteristics	Eukaryotic Cells	Prokaryotic Cells
Definition	Any cell that contains a clearly defined nucleus and membrane bound organelles	Any unicellular organism that does not contain a membrane bound nucleus or organelles
Examples	Animal, plant, fungi, and protist cells	Bacteria and Archaea
Nucleus	Present (membrane bound)	Absent (nucleoid region)
Cell Size	Large (10-100 micrometers)	Small (less than a micrometer to 5 micrometers)
DNA Replication	Highly regulated with selective origins and sequences	Replicates entire genome at once
Organism Type	Usually multicellular	Unicellular
Chromosomes	More than one	One long single loop of DNA and plasmids
Ribosomes	Large	Small
Growth Rate/Generation Time	Slower	Faster
Organelles	Present	Absent
Ability to Store Hereditary Information	All eukaryotes have this ability	All prokaryotes have this ability
Cell Wall	Simple: Present in plants and fungi	Complex: Present in all prokaryotes
Plasma Membrane	Present	Present
Cytoplasm	Present	Present

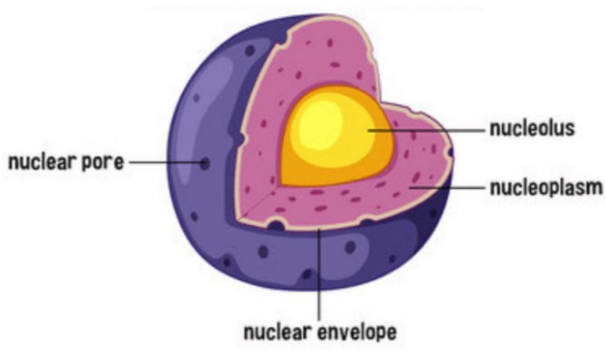
* Eukaryotic Cell



* A Cell



* Nucleus



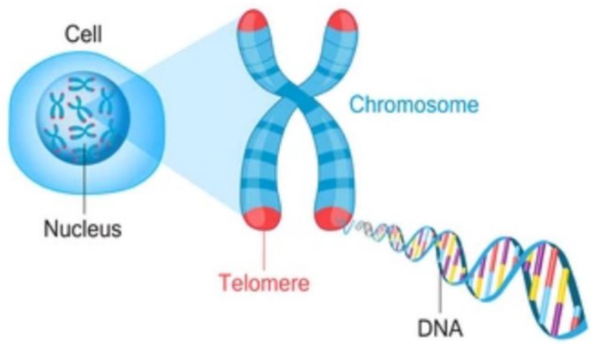
→ Function :

- The control centre of cell.
- Stores DNA.
- Makes RNA and protein.
- Makes ribosomes (nucleolus).

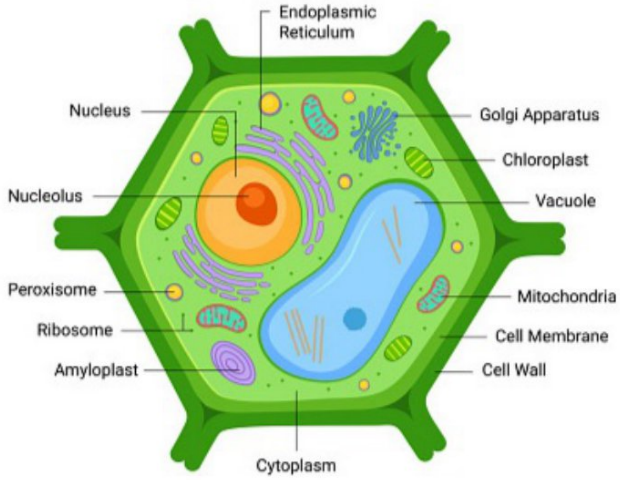
→ Structure :

- Double Membrane (Nuclear Envelope)

Cell
↓
Nucleus
↓
Chromosome
↓
contain information for inheritance from parents in the form of DNA.



* Cell Wall



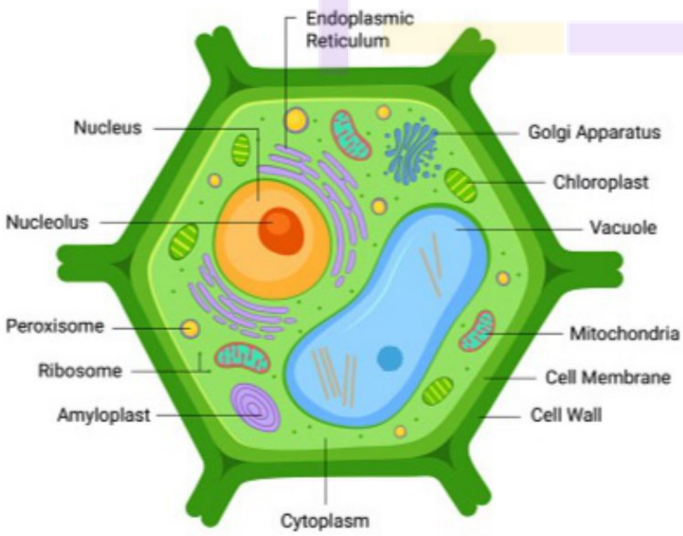
→ **Function :**

- Provides support and protection to the cell.

→ **Structure :**

- Lies outside cell membrane.
- Not present in animal cells.

* Cell Membrane



→ **Function:**

- Physical barrier, seperates internal environment, selective permeability.

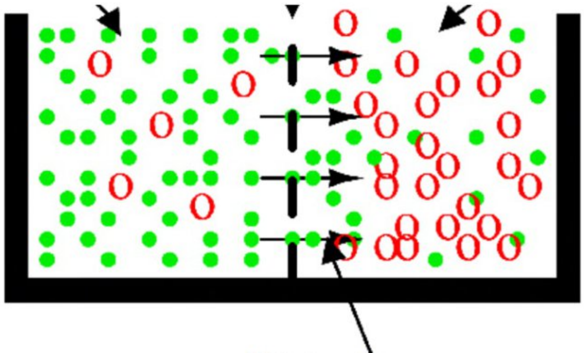
→ **Structure:**

- Semipermeable

* Semi Permeable Membrane

- Allows movement of only some particles.

• Movement from high concentration of solute to low concentration of solute.



* Osmosis V/s Diffusion

→ Osmosis :

• The movement of water molecules through such a selectively permeable membrane is called Osmosis.

• Osmosis is the passage of water from a region of high water concentration through a selectively permeable membrane to a region of low water concentration till equilibrium is reached (matlab jab tak dono side concentration same nahi ho jata).

→ Diffusion :

• Diffusion is defined as the movement of individual molecules of a substance through a semipermeable barrier/

membrane from an area of higher concentration to an area of lower concentration.

1. If the medium surrounding the cell has a higher water concentration than the cell, meaning that the outside solution is very dilute, the cell will gain water by osmosis.
   - Such solution is known as a hypotonic solution.

- Water molecules are free to pass across the cell membrane in both directions, but more water will come into the cell than will leave.
  - The net (overall) result is that water enters the cell.
  - The cell is likely to get swell up.

Endosmosis

2. If the medium has exactly the same water concentration as the cell, there will be no net movement of water across the cell membrane.
   - Such a solution is known as an isotonic solution.

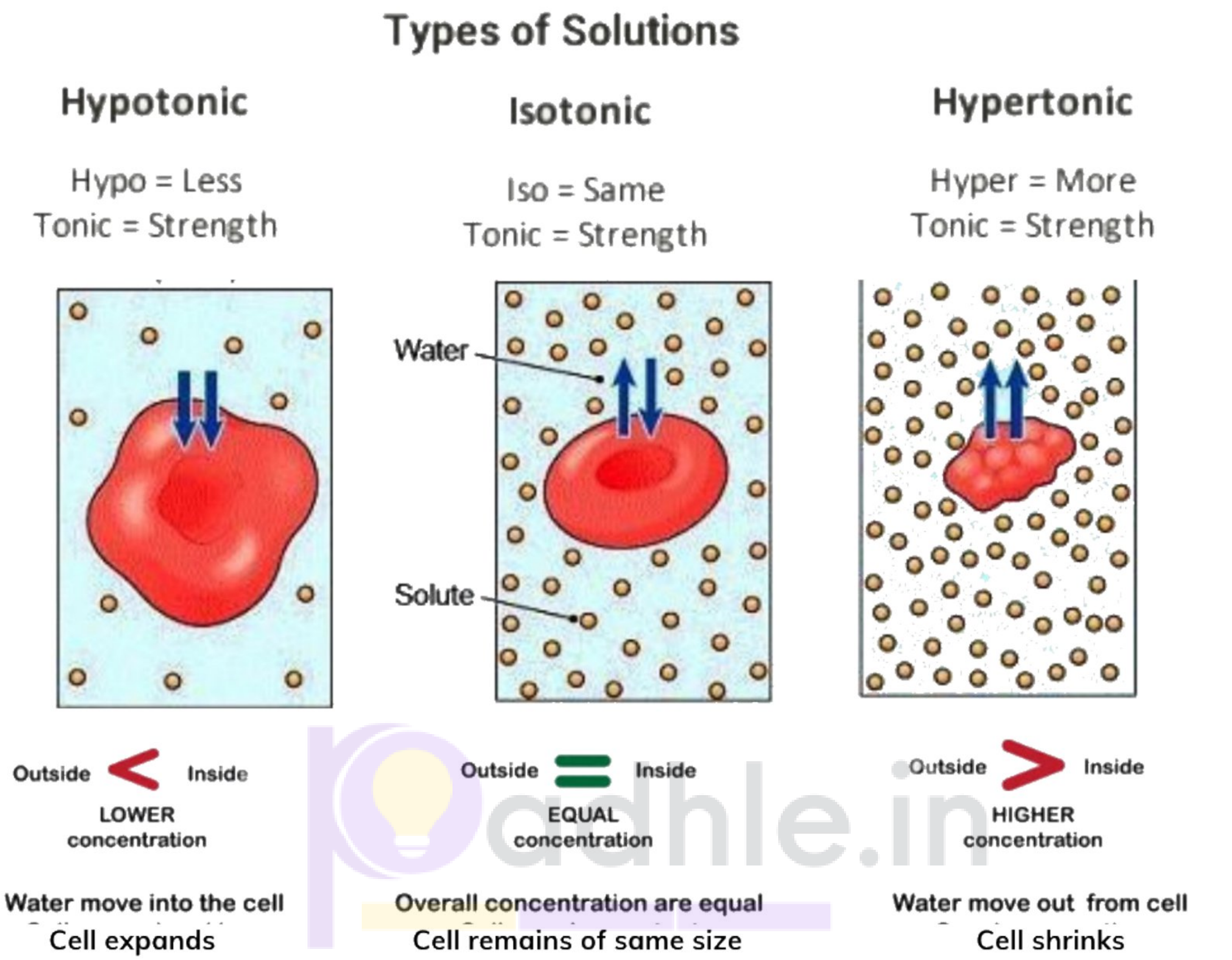
- Water crosses the cell membrane in both directions, but the amount going in is the same as the amount going out, so there is no overall water movement.
  - The cell will stay the same size.

Equilibrium

3. If the medium has a lower concentration of water than the cell, meaning that it is a very concentrated solution, the cell will lose water by osmosis.
   - Such a solution is known as a hypertonic solution.

- Again, water crosses the cell membrane in both directions, but this time more water leaves the cell than ensure it.
  - Therefore the cell will shrink.

Exosmosis



- Movement from high concentration to low concentration.

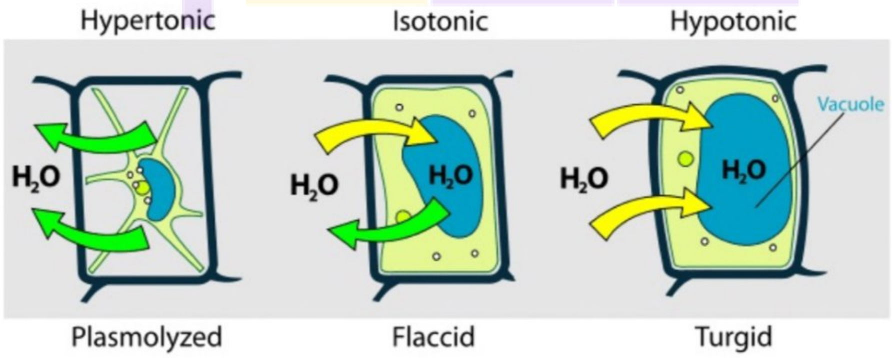
* Cytoplasm

→ Function :

- Organelles float in cytoplasm, helps in movement.

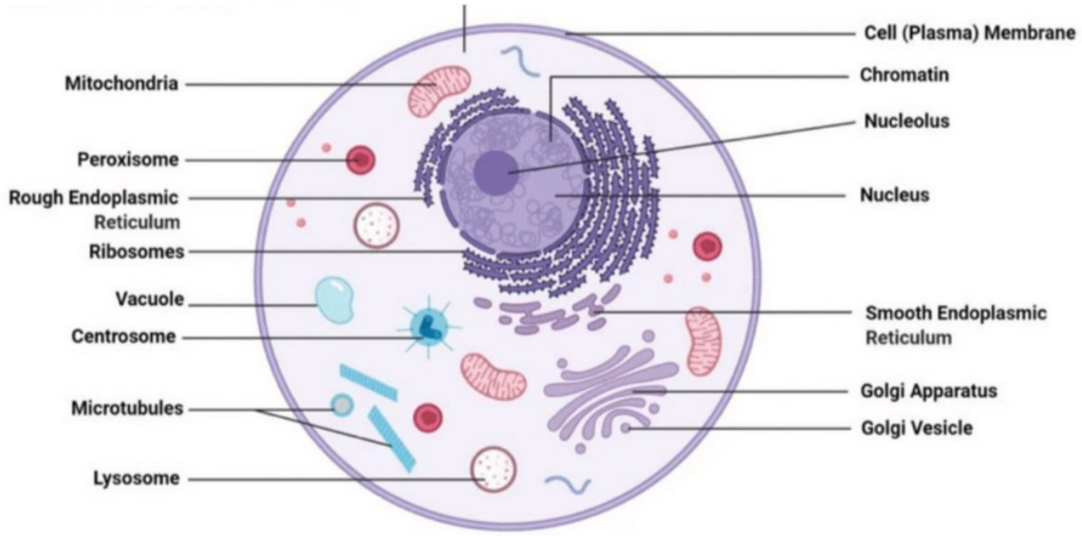
→ Structure :

- Fluid inside cell membrane.
- Contains water, salts, organic compounds.



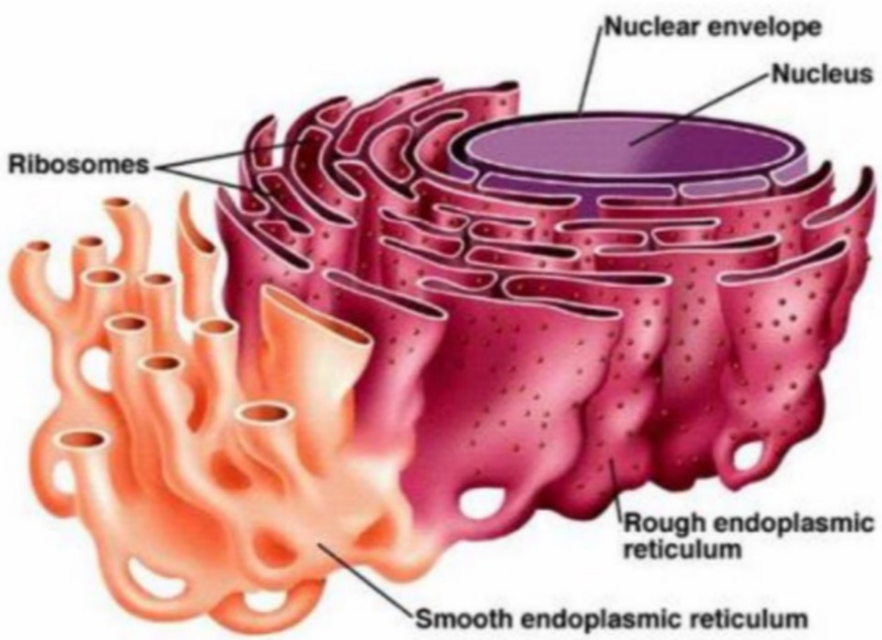
- Plasmolysis is the process of shrinkage or contraction of the protoplasm of a plant cell as a result of loss of water from the cell.
- Plasmolysis is one of the results of osmosis and occurs very rarely in nature, but it happens in some extreme conditions.

* # Cell Organelles



* # Endoplasmic Reticulum

- 'Post - Manufacturing Unit'



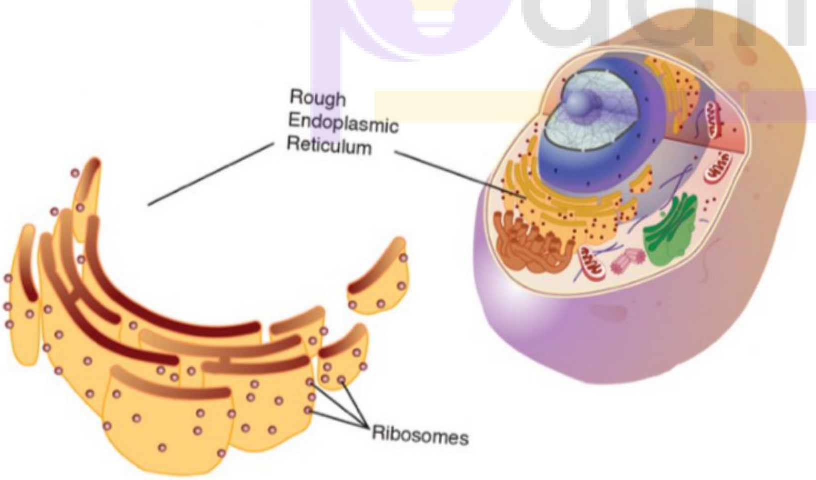
→ Structure :

- ER is a large network of membrane-bound tubes and sheets.
  - Long tubule type shape.

→ Function :

- The endoplasmic reticulum can either be smooth or rough, and in general its function is to produce proteins for the rest of the cell function.

* Rough ER



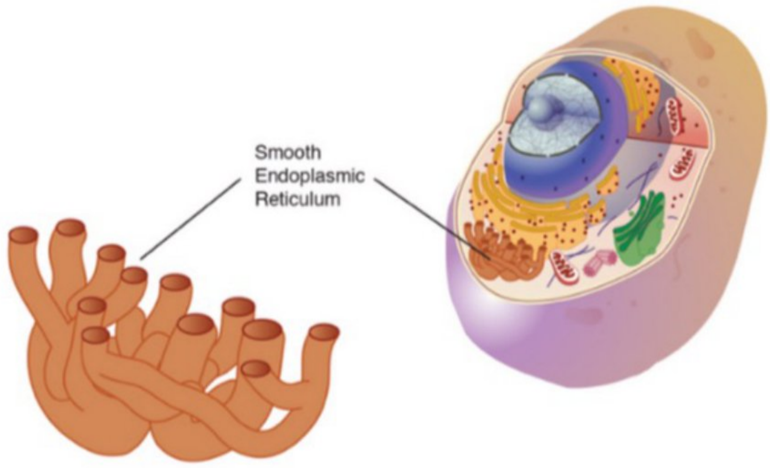
→ Structure :

- Rough, has ribosomes.

→ Function:

- Ribosomes are sites of protein manufacturing.

* # Smooth ER



→ Structure :

- Smooth

→ Functions :

- Manufacturing fat molecules, lipids, etc.

* # Membrane Biogenesis

- Process of building the cell membrane from these proteins and lipids.

- Some other proteins and lipids are used in formation of enzymes and hormones.

- SER of liver cells of vertebrates help in detoxifying any toxic substance.

* Golgi Apparatus

- 'Post - Transport and packer'

→ Structure :

- Consists of a system of membrane - bound vesicles (flattered sacs) arranged approximately parallel to each other in stacks called cristerns.

- These membranes often have connections with the membranes of ER and therefore constitute another portion of a complex cellular membrane system.

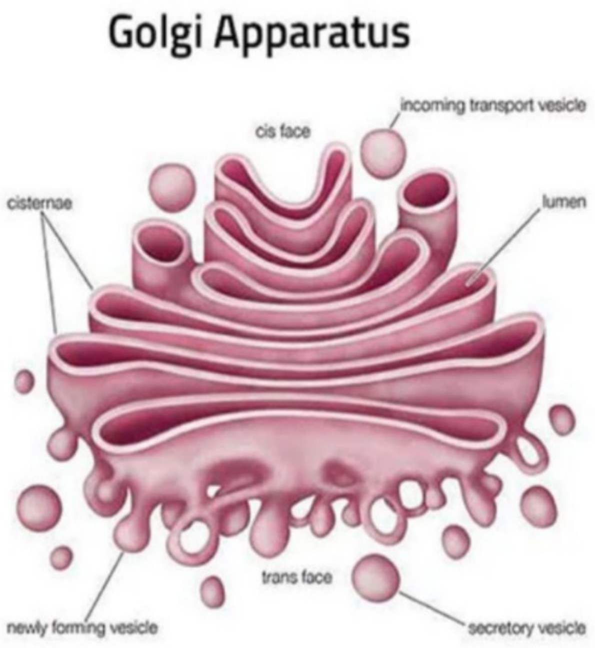
→ Functions :

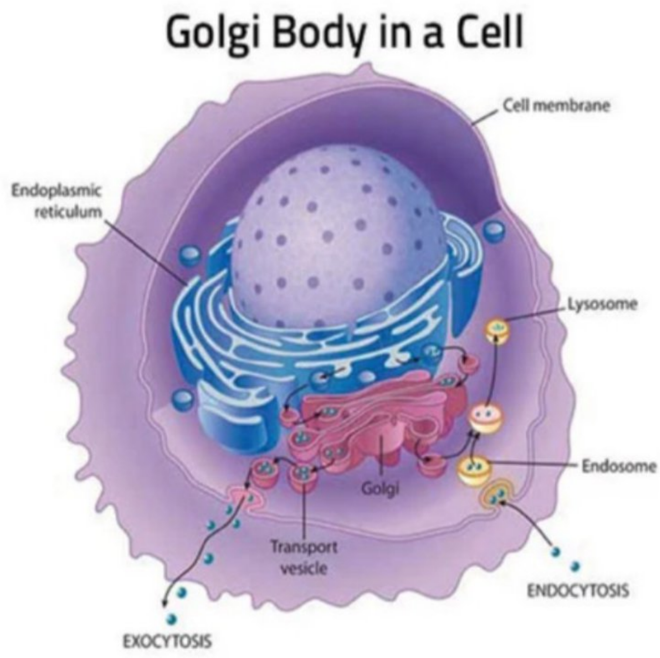
- The material synthesized near the ER is packaged and and dispatched to various targets inside and outside the cell through the Golgi apparatus.

- Its functions include the storage, modification and packaging of products in vesicles.

- In some cases, complex sugars may be made from simple sugars in the Golgi apparatus.

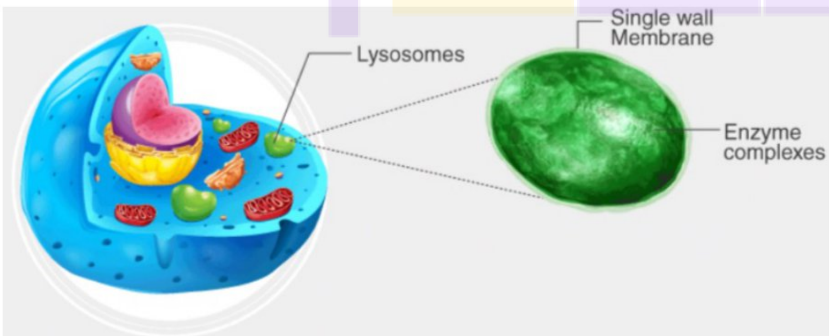
- The Golgi apparatus is also involved in the formation of lysosomes.





* Lysosomes

- 'Post - Sucidal Bags'



→ Structure :

- Lysosomes are membrane - bound sacs filled with digestive enzymes.

→ Function :

- Lysosomes are kind of waste disposal system of the cell.
- These help to keep the cell clean by digesting any foreign material as well as worn-out cell organelles.

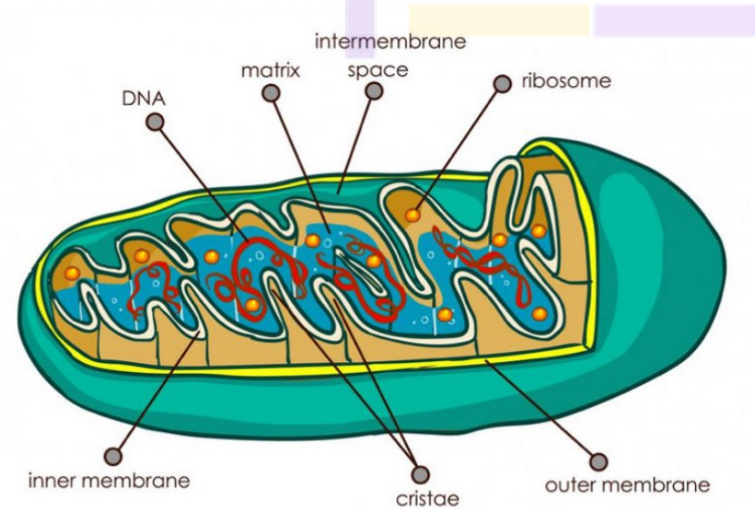
- Lysosomes are able to do this because they contain powerful digestive enzymes capable of breaking down all organic material.

- During the disturbance in cellular metabolism for eg., when the cell gets damaged, lysosomes may burst and the enzymes digest their own cell.

- Therefore, lysosomes are also known as the 'suicide bags' of a cell.

* Mitochondria

- 'Post - Powerhouse of the cell'



→ Structure :

- Mitochondria have two membrane coverings.

- The outer membrane is porous while the inner membrane is deeply folded.

→ Function :

- The energy required for various chemical activities needed for life is released by mitochondria in the form of ATP (Adenosine Triphosphate) molecules.

- ATP is known as the energy currency of the cell.
- The body uses energy stored in ATP for making new chemical compounds and for mechanical work.
- Mitochondria are strange organelles in the sense that they have their own DNA and ribosomes.
- Therefore, mitochondria are able to make some of their own proteins.

* Plastids

– 'Post – Kitchen of the cell'

→ Structure :

- Plastids are present only in plant cells.
- The internal organisation of the Chloroplast consists of numerous membrane layers embedded in a material called the stroma.
- These are similar to mitochondria in external structure.
- Like the mitochondria, plastids also have their own DNA and ribosomes.

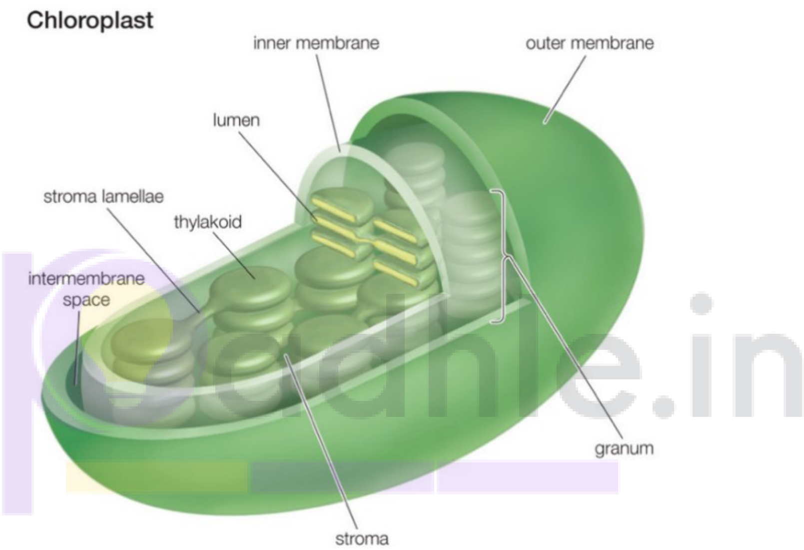
Plastids

Chromoplast (coloured plastids)

Chromoplasts containing the pigment chlorophyll are known as chloroplasts. Chloroplasts are important for photosynthesis in plants.

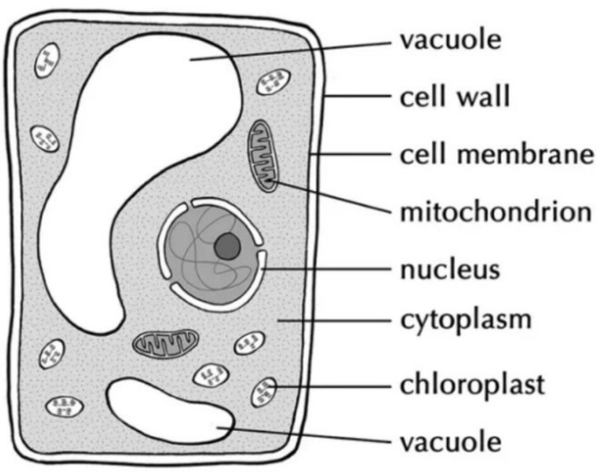
Leucoplasts (white or colourless plastids)

Leucoplasts are primarily organelles in which materials such as starch, oils and protein granules are stored.



* Vascuole

- 'Post - Storehouse of the cell'



→ <u>Structure</u> :

• Vacuoles are small-sized in animal cells while animal cells have very large vacuoles.

- The central vacuole of some plant cells may occupy 50-90% of the cell volume.

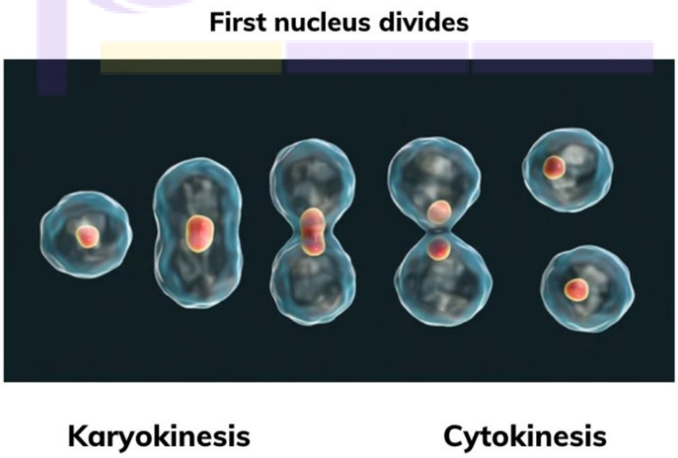
- In plant cells vacuoles are full of cell sap and provide turgidity and rigidity to the cell.

→ <u>Function</u> :

- Vacuoles are storage sacs for solid or liquid contents.

- Amino acids, sugars, various organic acids, some proteins and important substances in the life of plant cell are stored.

* Cell Division



Mitosis and Meiosis